communications physics

A Nature Portfolio journal

Article

https://doi.org/10.1038/s42005-026-02841-y

Electrical conductivity structure drives anomalous and potentially time-varying magnetic fields at Uranus and Neptune

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Uranus and Neptune exhibit magnetic fields that are non-dipolar and non-axisymmetric, unlike those of any other known planet. These fields are generated by fluid motions in electrically conducting planetary interiors via dynamo action, offering a window into otherwise inaccessible deep processes. Understanding why Uranus and Neptune have such unusual magnetic fields provides critical insight into dynamo processes across planets and stars. Here, we present numerical dynamo models that produce large-scale magnetic features consistent with the observed fields of Uranus and Neptune, with the radial profile of electrical conductivity — controlled by interior composition and temperature — being a critical ingredient. These non-dipolar dynamos exhibit substantial temporal variation, although the rate fluctuates and can be transiently slow. Changes in the intensity and/or orientation of Uranus' and Neptune's magnetic fields are therefore plausible over more than 30 years since the *Voyager 2* flybys, though the extent of any such variation remains poorly constrained observationally as each planet has been measured in situ only once. This potential secular variation can affect the interpretation of auroral observations and inform planning for future missions to the Uranian and Neptunian systems.

The *Voyager 2* flybys revealed that Uranus and Neptune, traditionally referred to as the ice giants, possess intrinsic magnetic fields generated within their interiors[1,2]. However, the characteristics of these fields overturned the conventional picture of a planetary dynamo dominated by an axial dipole. Ice giant magnetic fields are highly non-dipolar: the axial dipole accounts for only 12% of the total surface magnetic energy[3–6], whereas all other known planetary dynamos produce dipole-dominated fields, with the axial dipole contributing 74–98% of the total observed surface magnetic energy[7–10] (for details, see Supplementary Note 1).

Uranus' and Neptune's magnetic fields are also strongly non-axisymmetric, exhibiting pronounced power in the spherical harmonic (SH) order $m = 1$ in the surface magnetic power spectra[3–6]. In contrast, the $m = 1$ term is an order of magnitude weaker than the axisymmetric $m = 0$ component at Jupiter and the Earth, more than three orders of magnitude weaker at Mercury, and remains consistent with zero at Saturn[9,11] (Supplementary Table 1). Because the *Voyager 2* flybys provided only single snapshots in time, it is unlikely that both encounters happened to capture transient magnetic features at both planets; the persistent $m = 1$ peak—hereafter referred to as the "M1 peak"—at both planets thus appears to be an intrinsic characteristic of the ice giant magnetic fields.

A direct consequence of these complex spatial features is that Uranus' and Neptune's magnetic fields may be time-variable, as dynamo simulations have shown that non-dipolar solutions tend to have strong temporal variations, with smaller-scale magnetic structures evolving more rapidly than large-scale components[12–14]. As their magnetic fields have only been measured in situ once, it is possible that these fields have changed in amplitude and/or orientation since the *Voyager 2* observations in the late 1980s. Such variations on decadal or shorter timescales may introduce systematic uncertainties in interpreting auroral features and atmospheric emission diagnostics via telescopic observations. In particular, recent detections of the molecular ion H_3^+ at both planets, observed with the *Keck II* telescope and the *James Webb Space Telescope (JWST)*, have been interpreted using magnetic field models derived from the *Voyager 2* data[15,16]. Similarly, recent determination of Uranus' rotation rate using images of ultraviolet aurorae from the *Hubble Space Telescope (HST)* assumed that its magnetic field has remained static since 1986[17,18]. Recent auroral observations have begun to address secular variation (SV) through the inferred positions of emissions.

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At Uranus, aurorae observed by *Keck-NIRSPEC* and *IRTF-iSHELL*[19], *HST*[20], and *JWST*[21] are found at broadly similar locations to those from the *Voyager 2* flyby [6], with approximately 10–20° of latitudinal uncertainty. At Neptune, however, the southern auroral emission appears to shift latitudinally by ~30° between 1989 and 2023[15], though whether this reflects secular variation of the magnetic field remains uncertain.

Clues to the origin and behavior of Uranus' and Neptune's magnetic fields may lie in their energy budgets and internal structures, since their dynamos require convection of an electrically conducting fluid[22]. The two planets exhibit drastically different intrinsic heat fluxes: Uranus emits little excess heat beyond absorbed sunlight, whereas Neptune emits nearly twice as much energy as it receives from the Sun[23–26]. Yet the two planets maintain similarly complex, non-dipolar, non-axisymmetric magnetic fields with similar field strength. Reconciling this apparent decoupling between thermal output and field morphology requires understanding how electrical conductivity and convective geometry shape the regions where magnetic fields are generated.

Gravitational data from *Voyager 2*, resolved up to the J_4 moments, have been used to infer internal composition and density profiles ([27] and references therein). However, due to the inherent degeneracy of gravity field inversions, a range of interior models—with varying water-to-rock ratios and temperatures—can equally satisfy the gravitational constraints. Classical models envision a hydrogen-rich molecular envelope overlying a water-dominated mantle and a central rocky core, but more recent studies hypothesize that the mantle is separated into a water-rich layer above a polymeric C-N-H fluid[28] and that there may be substantial mixing between rock and volatile components[29–32].

The poorly constrained interior temperatures and compositions lead to large uncertainty in the electrical conductivity profiles of Uranus and Neptune. High-pressure experiments and ab initio simulations show that water transitions from molecular to ionic phases at ~30–50 GPa, reaching conductivities at or above 2×10^3 S/m at depths of ~0.7–0.8 R—the region traditionally assumed to host dynamo action[33–38]. However, the deep interiors may deviate from a pure-water composition[39], incorporate substantial rock mixing[40], and/or transition to a superionic phase with markedly different electrical and mechanical properties[41–45]. This uncertainty introduces fundamental ambiguity in the location, efficiency, and nature of dynamo generation.

Various mechanisms have been proposed to explain the non-dipolar, non-axisymmetric magnetic fields of Uranus and Neptune (see recent reviews by refs. 22,46). Early work invoked deeply embedded dynamos, but they are difficult to reconcile with the strong non-dipolar features observed at planetary surfaces because smaller-scale field components decay exponentially with distance from the source region[47]. Dynamo models with a deep stably stratified fluid layer below the convective dynamo region have been shown to produce ice-giant-like multipolar fields, provided the dynamo operates in a moderately thin shell geometry ($0.5 \le \chi \le 0.7$)[48,49]. These models assume the same electrical conductivity in both the stable and convective layers, though whether such a stable layer exists depends on the mechanical behavior of superionic ice at depth, which remains debated[42,43]. Alternatively, dynamo models with a large solid inner core ($\chi = 0.7$) that is less electrically conducting than the overlying convective region can also generate multipolar magnetic fields[49], though superionic water is expected to be more electrically conducting than ionic water[41], complicating this assumption. Both models produce relatively strong dynamos (as measured by the dimensionless Elsasser number, $\Lambda \sim 1$), substantially exceeding estimated ice giant values ($\Lambda \sim 10^{-4}$)[50]. Furthermore, multipolar fields can emerge in both thin ($\chi = 0.75$) and thick ($\chi = 0.35$) shell geometries when convection is sufficiently vigorous that the rotational constraints become weak[51]. These models produce field strengths ($\Lambda \sim 10^{-3}$) much closer to ice giant estimates than the thin-shell models above. However, this mechanism requires strong buoyancy forcing in the dynamo region, which may be difficult to sustain in the ice giants—particularly Uranus, given its weak internal heat flux and inferred limited convective vigor[23]. All of these dynamo models produce highly time-variable magnetic fields, yet none assess the temporal variation of the surface radial magnetic field.

What has not been considered in previous ice giant dynamo models is how interior properties—particularly the radial variation of electrical conductivity—affect magnetic field generation. Dynamo models of Jupiter and Saturn that incorporate radially variable conductivity profiles have shown that field morphology sensitively depends on the conductivity drop-off depths[52–54], and further affects dynamics in the proposed stably stratified layers for gas giant planets[55,56]. Similar behavior is expected to occur in Uranus and Neptune, where the electrical conductivity of water gradually increases with depth due to increasing pressure, a robust feature demonstrated across ab initio and high-pressure, high-temperature experiments[33–36,57]. We therefore use numerical dynamo simulations to investigate how radial variations in electrical conductivity, $\sigma(r)$, together with the geometry of the convective shell, χ, control the morphology and temporal variation of magnetic fields generated in ice-giant-like interiors (see "Methods" for details). Tables 1 and 2 list the control parameters and characteristic diagnostics, respectively, of the numerical simulations investigated here.

Figure 1 illustrates the corresponding profiles for all models. The radially variable conductivity profiles are characterized by the transition radius r_m, the conductivity value σ_m, and the decay parameter a, which together determine the radial location and steepness of the conductivity drop; the full functional form is given in the "Methods" section. Specifically, σ_3 in the thick shell geometry ($\chi = 0.35$) models the radially variable electrical conductivity adopted from ref. 33, using interior models from ref. 37. σ_1 and σ_2 are representative cases included to test the sensitivity to the depth and steepness of the conductivity drop. In thinner shells ($\chi = 0.5, 0.6$), the same relative conductivity-profile shapes are retained within the convective shell to isolate the effects of radial conductivity variation from those of shell geometry.

Our results identify radially varying electrical conductivity as a pathway to generating Uranus- and Neptune-like magnetic fields. We demonstrate that dynamo models incorporating an electrical conductivity that increases with depth can promote non-dipolar, non-axisymmetric magnetic fields, including the characteristic M1 peak observed in the magnetic power spectra of Uranus and Neptune. We further find that these ice-giant-like dynamos are intrinsically time-varying, although intervals of slow secular variation can also occur.

Table 1 | Control parameters of the dynamo simulations

Model	χ	$\sigma(r)$	E	Ra
0	0.35	σ_0 = (–, –, –)	1.0×10^{-4}	2.8×10^7
1	0.35	σ_1 = (0.813, 0.7, 10)	1.0×10^{-4}	2.8×10^7
2	0.35	σ_2 = (0.875, 0.2, 4)	1.0×10^{-4}	2.8×10^7
3	0.35	σ_3 = (0.817, 0.155, 1.345)	1.0×10^{-4}	2.8×10^7
4	0.5	σ_0 = (–, –, –)	1.0×10^{-4}	2.35×10^7
5	0.5	σ_1 = (0.856, 0.7, 10)	1.0×10^{-4}	2.35×10^7
6	0.5	σ_2 = (0.904, 0.2, 4)	1.0×10^{-4}	2.35×10^7
7	0.5	σ_3 = (0.859, 0.155, 1.345)	1.0×10^{-4}	2.8×10^7
8	0.6	σ_0 = (–, –, –)	1.0×10^{-4}	2.1×10^7
9	0.6	σ_1 = (0.885, 0.7, 10)	1.0×10^{-4}	2.1×10^7
10	0.6	σ_2 = (0.923, 0.2, 4)	1.0×10^{-4}	2.1×10^7
11	0.6	σ_3 = (0.887, 0.155, 1.345)	1.0×10^{-4}	2.1×10^7

$\chi = r_i/r_o$ is the aspect ratio of the dynamo shell. $\sigma(r) = (r_m, \sigma_m, a)$ defines the radial electrical conductivity profile (see "Methods" section), with σ_0 being uniform in radius. Ra describes the convective forcing, chosen such that models with different χ have comparable supercriticality ($Ra/Ra_c \sim 28$; Model 7 was run at a somewhat higher forcing). Other non-dimensional parameters are fixed at Ekman number $E = 10^{-4}$, Prandtl number $Pr = 1$, and magnetic Prandtl number $Pm_i = 5$.

Table 2 | Diagnostic parameters

Model	Rm	Ro_l	Λ	f_{M1}	$p_{M>1}$	$p_{l>1}$	tilt (°)	p^U_{tilt}	p^N_{tilt}
0	1389 ± 31	0.06 ± 0.01	2.2 ± 0.5	0.08	0%	0%	4 ± 2	0%	0%
1	741 ± 43	0.10 ± 0.01	(1.0 ± 0.3) × 10^{-2}	0.5	21.7%	37.1%	101 ± 63	4.1%	7.2%
2	647 ± 33	0.08 ± 0.01	(1.9 ± 0.6) × 10^{-4}	1.9	58.2%	72.7%	114 ± 44	13.5%	13.0%
3	470 ± 45	0.07 ± 0.01	(2.1 ± 1.0) × 10^{-4}	0.5	25.6%	46.2%	93 ± 51	14.4%	17.7%
4	694 ± 74	0.10 ± 0.01	5.0 ± 1.4	0.12	0%	0%	8 ± 5	0%	0%
5	393 ± 15	0.17 ± 0.01	(1.5 ± 0.3) × 10^{-2}	1.4	74.9%	71.8%	107 ± 52	9.3%	13.1%
6	278 ± 9	0.11 ± 0.01	(1.5 ± 0.3) × 10^{-4}	3.2	89.6%	89.7%	87 ± 50	13.0%	12.5%
7	275 ± 14	0.13 ± 0.01	(4.3 ± 1.3) × 10^{-4}	1.5	66.9%	56.7%	79 ± 40	16.9%	17.7%
8	1082 ± 52	0.22 ± 0.01	1.8 ± 0.3	0.13	0%	0%	5 ± 2	0%	0%
9	235 ± 7	0.24 ± 0.01	(1.8 ± 0.3) × 10^{-2}	1.9	82.6%	94.9%	105 ± 47	11.6%	8.8%
10	201 ± 6	0.20 ± 0.01	(1.8 ± 0.4) × 10^{-4}	3.0	91.4%	85.5%	106 ± 36	8.7%	8.0%
11	169 ± 6	0.18 ± 0.01	(4.2 ± 1.0) × 10^{-4}	2.1	78.1%	62.8%	83 ± 40	12.6%	12.6%

Rm is the volume-averaged magnetic Reynolds number, Ro_l is the local Rossby number, and Λ is the Elsasser number measured at the dynamo surface. f_{M1} is the relative $M1$ dominance in magnetic power assuming a dynamo surface at $0.8R$, $p_{M>1}$ is the percentage of snapshots in which the azimuthal order $m = 1$ power exceeds the axisymmetric $m = 0$ power, and $p_{l>1}$ is the percentage of snapshots in which power at degree $l = 2$ or $l = 3$ exceeds the dipole ($l = 1$). Tilt is the angle between the magnetic dipole and rotation axes, while p^U_{tilt} and p^N_{tilt} are the percentages of snapshots with dipole tilt within ±10° of the *Voyager 2* observed values for Uranus (59°) and Neptune (47°), respectively. These Rm, Ro_l, Λ, and tilt values are averaged over one viscous diffusion time (0.2 magnetic diffusion times) after the simulations have reached equilibrium, with the standard deviations also reported.

Results

Effects of variable electrical conductivity

We first consider models with constant electrical conductivity (σ_0) across three shell geometries ($\chi = 0.35, 0.5, 0.6$). All σ_0 models (Models 0, 4, and 8; Table 1) produce strong axial dipole-dominated magnetic fields at the top of the dynamo region (Fig. 2a, b, $1.0R$), with Elsasser numbers of $\Lambda \sim 2$–5. For each σ_0 model, we extrapolate the magnetic power spectra to the planetary surface assuming source radii of $0.8R$ (a common assumption for ice giants) and $1.0R$ (an unlikely end-member that provides an upper bound on high-degree surface power). Because potential-field attenuation scales with SH degree as r^{-2l-4}, extrapolating from a shallower source depth preferentially enhances the relative power at higher l. At $1.0R$, this flattens power at degrees above the dipole ($2 \le l \le 6$; Fig. 2a). Nevertheless, even this end-member configuration fails to reproduce the observed spectral peak at $l > 1$ or the pronounced M1 peak. We quantify this latter feature using the M1 dominance, f_{M1}, defined as the ratio of magnetic power in the $m = 1$ mode to that in the axisymmetric $m = 0$ mode, with the power in each order m summed over all SH degrees satisfying $m \le l \le 6$ for the simulations (see "Methods" section), so that the M1 peak includes contributions from the dipole, quadrupole, octupole and higher-degree terms.

In contrast, once a radially variable electrical conductivity profile is introduced, the axial-dipole-dominated, non-reversing magnetic fields transition to multipolar, aperiodically reversing fields across all three convective shell thicknesses.

The resulting magnetic field strength, as measured by the Elsasser number Λ, decreases by two to four orders of magnitude, accompanied by a reduction in the magnetic Reynolds number, Rm, which measures the ratio of magnetic induction to diffusion (Table 2). Consistent with this magnetic weakening, the local Rossby number (Ro_l, ratio of inertial to Coriolis forces at the convective length scale) increases, indicating a stronger relative role of inertia. This $\sigma(r)$ variation therefore modifies the dynamo's force balance: decreased electrical conductivity near the outer boundary suppresses the Lorentz force, as reflected in the reduced magnetic power spectra (see Supplementary Figs. 1–3), allowing inertia and buoyancy to dominate the outer-boundary force balance and giving rise to more vigorous, smaller-scale convection in the outer shell. This behavior is consistent with previous numerical simulations, which have shown that the outer boundary force balance strongly shapes the global magnetic field morphology[54,55,58]. These models also transiently sample *Voyager 2*-like dipole tilts (Table 2); occurrence rates are highest for the σ_3 profile, followed by σ_2 and σ_1. Within each model, the Uranus-like and Neptune-like tilt rates are comparable, indicating that a single dynamo regime can transiently reproduce either.

The time-averaged magnetic spectra show a similar trend across shell geometries. In the thick dynamo shell ($\chi = 0.35$) cases, only Model 2 with the σ_2 profile produces an M1-dominated spectrum ($f_{M1} = 1.9$) with enhanced $l = 2$ power, whereas the σ_1 and σ_3 cases have weaker M1 features ($f_{M1} \approx 0.5$; Supplementary Fig. 4). As the dynamo shell becomes thinner, the spectra generally become more multipolar and non-axisymmetric, with stronger M1 peaks in the variable-conductivity models (Supplementary Figs. 5–7); snapshot rates of non-dipolarity $p_{l>1}$ and the M1 peak $p_{M>1}$ rise in parallel, with σ_2 models reaching the highest M1 occurrence in every shell geometry. Supplementary Fig. 8 shows that variable electrical conductivity leads to substantially higher power in the low-frequency range ($<10^2$) in SH order $m = 1$ compared to analogous constant-conductivity results, consistent with the strong and persistent M1 peak observed in the time-averaged power spectra. This indicates long-lived non-axisymmetric dynamics in the $m = 1$ mode.

Taken together, these results show that radially variable electrical conductivity is a key ingredient for producing multipolar, non-axisymmetric magnetic fields: constant-conductivity models remain strongly axial-dipole dominated, whereas variable-conductivity models develop a strong M1 peak and enhanced non-dipolar structure. Models with a σ_2 profile most effectively enhance the time-averaged M1 peak, while thinner dynamo shells further increase the multipolar and non-axisymmetric characteristics.

Ice-giant-like models from spatial morphology

Motivated by the conductivity-driven changes in magnetic morphology described above, we next identify which models most closely resemble the observed magnetic fields, as modeled by Herbert[6] (AH_5 for Uranus), Holme and Bloxham[5] (HB96 for both Uranus and Neptune), and Connerney et al.[4] (O_8 for Neptune). As detailed in the "Methods" section, this comparison is quantified by S_{tot}, which combines snapshot-based root mean square (RMS) morphology (S_{rms}) and time-averaged M1-dominance (S_{M1}) assessments. Lower S_{tot} values indicate closer resemblance to the observed fields (see Supplementary Tables 2 and 3). The remaining snapshot-occurrence diagnostics reported in Table 2 (i.e., $p_{l>1}$, $p_{M>1}$, p^U_{tilt} and p^N_{tilt}) are for description only and do not enter the evaluation of ice-giant-likeness.

Across the three estimates of dynamo depth ($A = r_o/R$, the outer radius of the dynamo region as a fraction of the planetary radius; $A = 0.7, 0.75$, and 0.8) and the two Uranus magnetic field models, the top-ranked models are

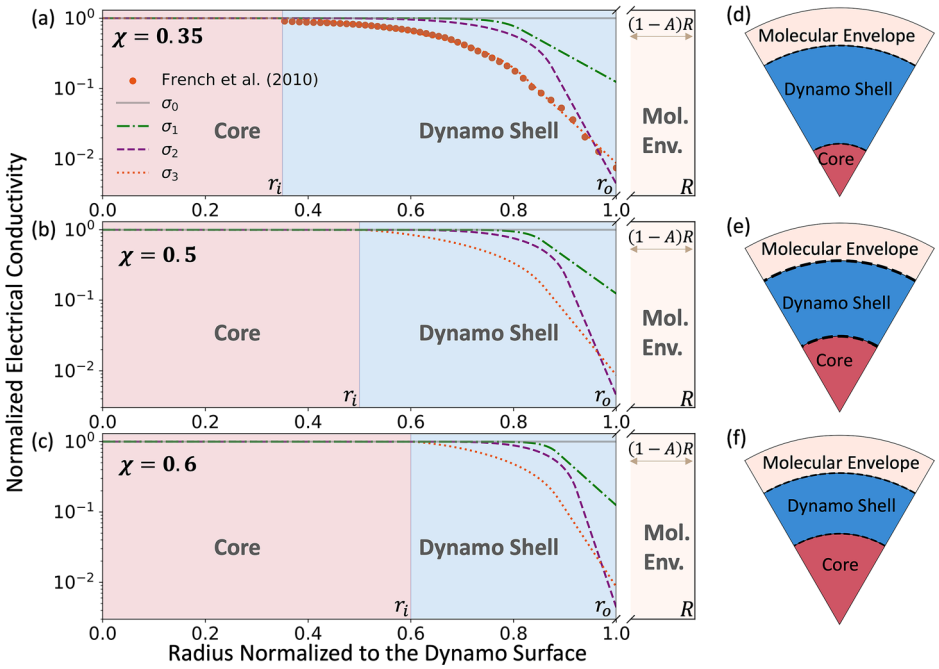


Fig. 1 | Schematic of Uranus' and Neptune's interior structure assumed for the dynamo models. a–c Constant (σ_0, gray solid) and radially variable electrical conductivity profiles (σ_1, green dash-dotted; σ_2, purple dashed; σ_3, orange dotted lines) for dynamo models with convective shell geometries of $\chi = r_i/r_o$ = 0.35, 0.5, and 0.6, shown in (**a**–**c**), respectively. Red dots in (**a**) represent the electrical conductivity data from ref.[33] with an interior profile from ref. 37. The dynamo shell is shown in blue, ranging from r_i to r_o with thickness $(1-\chi)AR$. The core, ranging from 0 to r_i with thickness χAR, is shown in pink. The molecular envelope (Mol. Env.) is shown in beige, ranging from r_o to R with thickness $(1-A)R$. **d**–**f** The associated spherical shell geometry illustrations for χ = 0.35, 0.5, and 0.6, respectively, shown with a fixed A = 0.8. Our simulations do not model the molecular envelope, which is assumed to be electrically insulating, and treat the core region as a freely rotating solid with uniform electrical conductivity for simplicity.

consistently Model 10 (χ = 0.6, σ_2), Model 6 (χ = 0.5, σ_2), and Model 11 (χ = 0.6, σ_3), although their relative order varies with the observational reference field (Supplementary Table 2). All three are relatively thin-shell models with variable conductivity profiles. The thick-shell models (χ = 0.35, Models 0–3) consistently rank low because their time-averaged M1 peaks fall below the observed level of dominance. The ranking among the top models is therefore governed primarily by how frequently and how closely individual snapshots resemble the observed Uranian field. On this basis, the σ_2 models (Models 6 and 10) produce the highest occurrence rates of Uranus-like snapshots, whereas Model 11 (σ_3) produces fewer but similarly well-matched snapshots.

For Neptune, the ranking is more sensitive to the choice of observational magnetic field model. When comparing against HB96, Model 9 (χ = 0.6, σ_1) ranks highest at all three dynamo depths, followed by Model 6 (χ = 0.5, σ_2) and Model 10 (χ = 0.6, σ_2). Relative to O_8, Model 11 (χ = 0.6, σ_3) ranks highest, again followed by Model 6 and Model 10. Model 7 (χ = 0.5, σ_3) enters the top three at the shallowest dynamo depth (A = 0.8) for both reference fields, and at A = 0.75 relative to O_8. In both cases, medium- to thin-shell models (χ = 0.5–0.6) dominate the top of the ranking, whereas thick-shell models again rank low because of their smaller f_{M1} values. This sensitivity to the observational reference field reflects differences between the HB96 and O_8 inversions, particularly in their inferred M1 structure, such that no single model is uniquely favored. Nevertheless, both references consistently identify variable-conductivity models at χ = 0.5–0.6, spanning the σ_1, σ_2, and σ_3 profiles, as the most successful at producing Neptune-like snapshots.

Ice-giant-like models from temporal variability

Planetary dynamos in the Solar System show diverse secular variation: the geomagnetic field undergoes reversals, excursions, westward drift, and sub-decadal "jerks"[59–61]. Jupiter's field evolved measurably between the *Pioneer 10* and *Juno* eras[8,62], whereas Saturn's remained extremely stable[63,64]. However, since the magnetic fields of Uranus and Neptune have each been measured in situ only once by the *Voyager 2* flybys, no temporal variation has been established unequivocally, although potential time variability in auroral emission locations[6,15,19–21] suggests that SV may be possible.

Figure 3 demonstrates that the most Uranus-like snapshot from Model 6 (panel c) and the most Neptune-like snapshot from Model 11 (panel d) can change substantially on decadal timescales, with their close morphological resemblance to the respective observed fields (panels a and b) deteriorating markedly after 32 years (panels e and f). Details on converting the non-dimensional simulated time to dimensional units are given in the "Methods" section.

Among the most Uranus-like models, both fast and slow secular variation can occur during their brief intervals of optimal field morphology (marked by shaded intervals in Fig. 4). The relevant comparison is therefore snapshot-to-snapshot: because each planet was observed at only a single epoch, we treat a model snapshot that transiently matches that epoch as a valid analog, and the morphology score ("Methods") rewards models that produce such low-misfit snapshots frequently. For Model 11 (Fig. 4g–l), the dipole tilt would shift by 84° over 40 years (t = 51–91 years). By contrast, for Model 6 (Fig. 4a–f), the dipole tilt variation over 40 years (t = 234–274 years) is projected to be 13°, consistent with the 10–20° latitudinal uncertainty for Uranus' aurorae.

Similarly, among the most Neptune-like cases shown in Supplementary Fig. 9 (both compared against the HB96 Neptune field model), Model 9 produces two brief Neptune-like intervals, the more pronounced of which shows a projected dipole tilt variation of about 11° over 40 years. By contrast, Model 11 exhibits several distinct time windows that satisfy Neptune-like conditions, with dipole tilt variations of 26°–56° over 40 years.

The differing SV timescales, as demonstrated by Fig. 4e, k, are linked to the relative dipole contribution. Periods of weaker SV coincide with intervals where the dipole component remains relatively strong and stable (Fig. 4c), similar to the observed Uranian field. Because the dipole evolves more slowly than the higher-degree components, these intervals exhibit slower overall secular variation. However, the slow SV shown in Fig. 4a–f is transient: the dipole SV timescale τ_{dip} decreases shortly after this brief interval, suggesting that the slow-SV state is temporary and can transition to faster SV.

Discussion

Non-dipolar magnetic fields with a pronounced M1 peak emerge across a range of parameter space when the electrical conductivity increases with depth. Such conductivity gradients are generically expected in planetary interiors due to pressure- and temperature-dependent material properties, such as the molecular-ionic transition of ices under Uranus and Neptune interior conditions[33–36,41,57]. The M1 peak is robust across all shell thicknesses tested, and thinner shells (χ = 0.5, 0.6) tend to produce ice-giant-like field

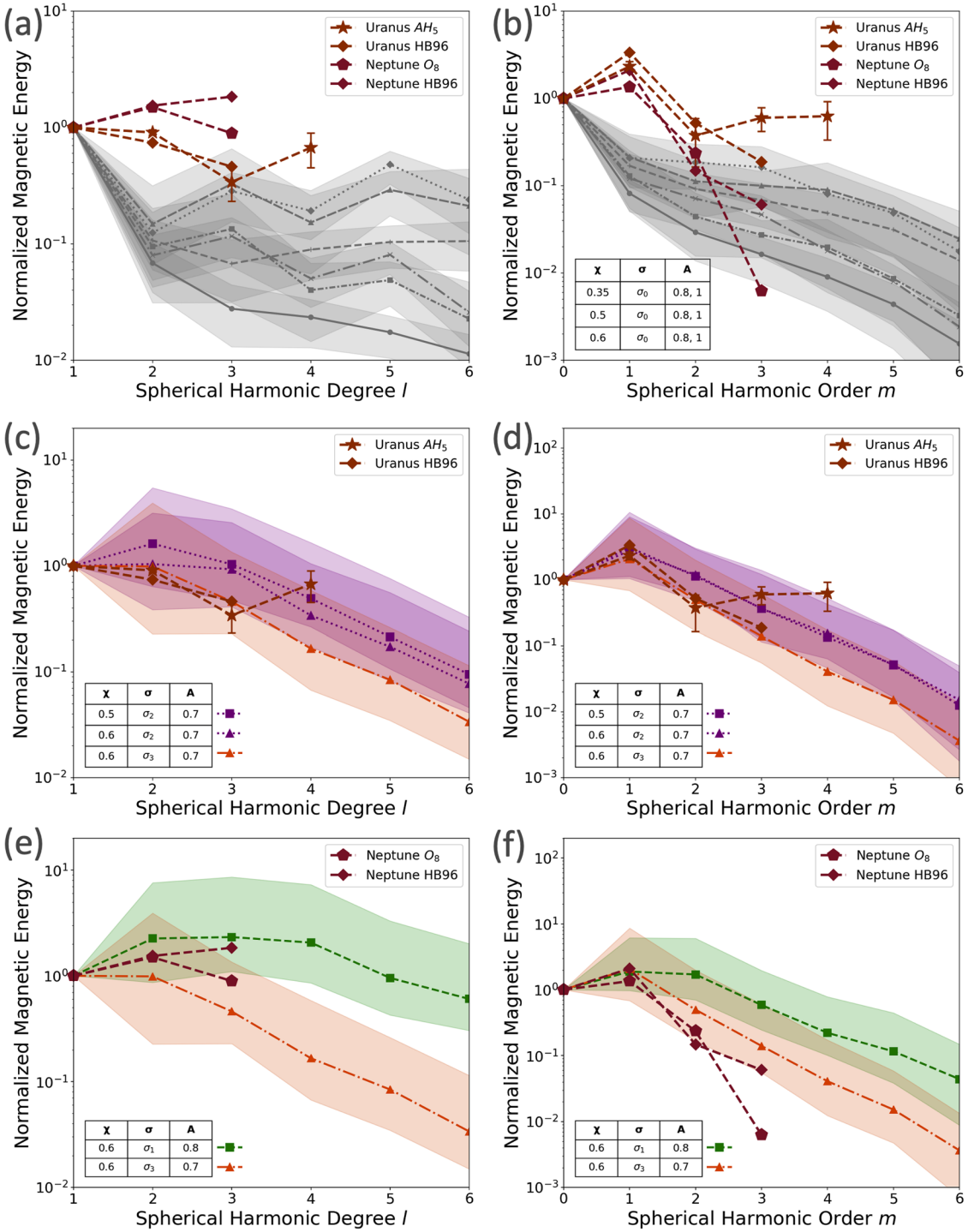


Fig. 2 | Magnetic power spectra. Magnetic power as a function of SH degree l (**a**, **c**, **e**) and order m (**b**, **d**, **f**) normalized, respectively, by power in the dipole $l = 1$ and the axisymmetric $m = 0$ mode. Burnt orange and crimson symbols denote the observed surface spectra of Uranus and Neptune, respectively, plotted at the 1-bar equatorial planetary radius ($1.0R$). Pentagrams indicate spectra from ref.[6] (error bars: propagated 1σ AH_5 coefficient uncertainties), pentagons from ref.[4], and diamonds from ref.[5]. Gray lines (**a**, **b**) show simulation spectra with constant electrical conductivity σ_0 and varying dynamo shell thickness χ, assuming dynamo depths of $0.8R$ and $1.0R$. **c**, **d** Show the top three models (Models 6, 10, 11) that most closely resemble the Uranus magnetic field morphology for both the HB96 and AH_5 field models. All models assume a dynamo depth of $0.7R$. **e**, **f** Present the magnetic power spectra of Models 9 and 11, which most closely resemble the Neptune magnetic field models HB96 and O_8, respectively. Model 9 (11) is plotted assuming a dynamo depth of $0.8R$ ($0.7R$). Simulation curves show the mean spectrum over the n consecutive snapshots spanning up to the final 0.1 magnetic diffusion time of each run after equilibration ($n = 1001$ for all models except Model 11, $n = 944$ and Model 8, $n = 740$), with same-color shaded bands marking the 1σ-equivalent interval (16th–84th percentile range) across these n snapshots. Spectra are shown to $l_{max} = 6$, with degree power summed over $m \le l$ and order power summed over $l \ge m$. Observed coefficients extend only to $l = 3$ (HB96, O_8) or $l = 4$ (AH_5) and are not compared beyond.

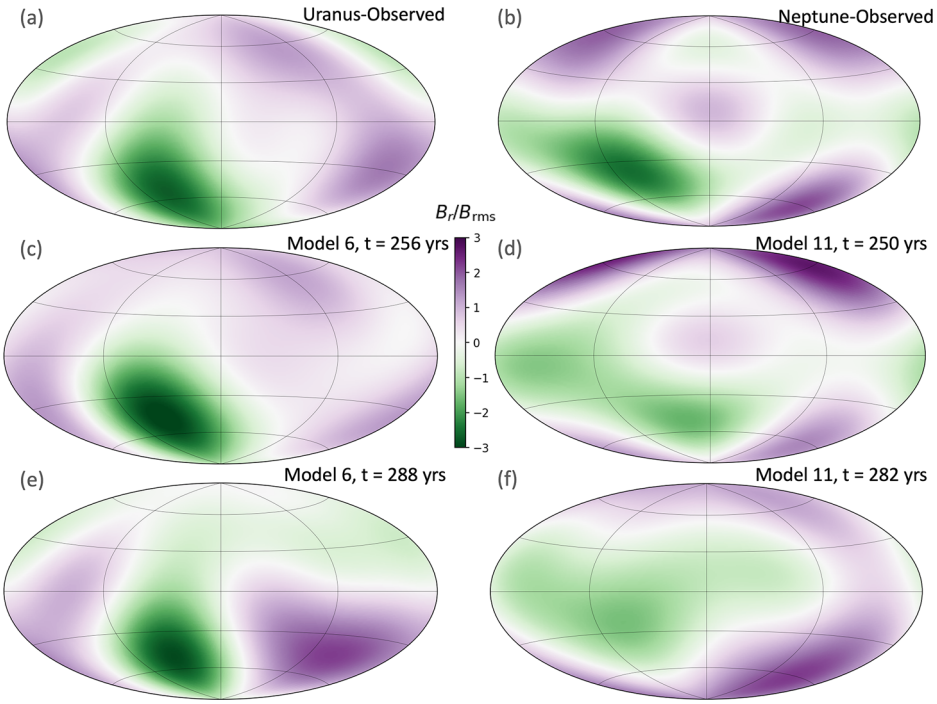


Fig. 3 | Radial magnetic fields at the planetary surfaces. Radial magnetic field B_r at the planetary surface in Hammer projection, normalized by the root-mean-square field strength $B_{\rm rms}$, with dynamo depth at $0.7R$ for Model 6 (Uranus) and $0.8R$ for Model 11 (Neptune). **a**, **b** Observed Uranus and Neptune fields[5], each with $l_{\rm max} = 3$. **c**, **e** Model 6 snapshot with the most Uranus-like morphology and 32 years later. **d**, **f** Model 11 snapshot at its closest Neptune-like morphology and 32 years later. Modeled fields are truncated at $l_{max} = 3$. Dimensional times use the magnetic diffusion time τ_λ ("Methods"; $\sigma = 10^4$ S/m), with $\tau_\lambda \approx 32$ kyr for Model 6 ($R = R_{\rm U}$) and ≈25 kyr for Model 11 ($R = R_{\rm N}$). Years reference the time series in Fig. 4 (Uranus) and Supplementary Fig. 9 (Neptune).

morphologies more frequently. We note that the M1 peak remains a non-unique characteristic and does not by itself define ice-giant-likeness. To test sensitivity to the dynamical regime, we considered two additional *Ra* values for a fixed $\chi = 0.35$ and σ_1 interior configuration (Models 13–14; Supplementary Table 4), finding that the magnetic field morphology shows a similarly weak dependence on supercriticality (Supplementary Fig. 10). Overall, these results indicate that the proposed pathway does not require finely tuned interior conditions, such that similar magnetic morphologies can arise across a range of shell geometries and dynamical regimes, even if Uranus and Neptune differ substantially in their interior structure, composition, or thermal evolution.

Although global dynamo simulations cannot reach the Ekman numbers (E, ratio of viscous to Coriolis forces) expected in planetary interiors[65], appropriate parameter scalings allow them to reproduce key dynamo diagnostics and force balances[66,67]. We extend Model 7 to a smaller Ekman number (Model 12, $E = 3 \times 10^{-5}$ compared to $E = 10^{-4}$) and find that the ice-giant-like characteristics, including the M1 peak, are sustained (Supplementary Fig. 11 and Supplementary Table 5), suggesting that this feature is dynamically robust within the accessible parameter range.

Our ice-giant-like dynamo models can produce Uranus-like snapshots with slow secular variation, but only transiently — a dipole tilt shift eventually occurs. If future observations show Uranus' slow secular variation to be a long-lived feature, this could indicate an electrical conductivity higher than currently assumed in the dynamo region, slowing magnetic diffusion. For example, extrapolating from Model 6's Uranus-like slow-SV interval (Fig. 4a–f), the dipole tilt would remain stable to within 5° change over ~500 years if the bulk electrical conductivity were one order of magnitude higher than the standard ionic water estimate ($\sigma = 10^5$ vs 10^4 S/m). This could suggest that Uranus' dynamo region is hotter than originally thought, leading to higher electrical conductivity (e.g., due to the presence of non-adiabatic regions[27,32]), and/or that the dynamo region contains materials with intrinsically higher electrical conductivity[40,68].

Future work should also consider radial variations in other interior properties, such as density structure, thermal conductivity, and stable layering, to determine how each influences the spatial structure of the magnetic field and its temporal variation. Several density scale heights are expected across the dynamo and semiconducting region[27,37,38], while ab initio studies suggest that thermal conductivity increases with depth in ice giant interiors[69,70]. Stable stratification in the deep interior is motivated by recent studies suggesting the deep interior may host viscous superionic ices or compositionally stratified layers[28,43,44]. These conditions may further modify convective vigor and the development of strong zonal flows, both of which can interact with and influence dynamo behavior. Separately, the atmospheric zonal winds observed at cloud level are constrained to be shallow—confined to a weather layer no more than ~ 1000 km deep[71–73]—and thus overlie and are largely decoupled from the dynamo region.

Despite current observational and computational limitations, our results indicate that ice-giant-like dynamos can, in some intervals, undergo substantial magnetic field variations on decadal timescales that, if confirmed by future in situ or auroral measurements, could have significant impacts on the surrounding planetary magnetospheres (see the reviews by refs. 74,75). These impacts include the interaction with the solar wind, where the changing planetary field strength and orientation just inside the magnetopause boundary is central to energy transport processes such as magnetic reconnection and Kelvin-Helmholtz instability[76–84]; this evolving interaction may be responsible for the long-term thermosphere cooling at both planets[85]. The radiation belts of both planets would also be affected: recent work has shown that relatively strong non-dipolar field components affect the drift of, and promote precipitation of, trapped energetic particles at Uranus—an effect that applies to Neptune as well, where changing Gauss coefficients will influence precipitation patterns onto each atmosphere[86–88]. Among the broader magnetospheric impacts are time-dependence in how energy cycles through the entire magnetospheric system[89–91], potential differences in electric current closure and auroral emissions across polar regions, and evolution of the spectrum of inducing magnetic fields around moons of both planets that may harbor subsurface oceans[92,93].

Methods

Fundamental equations and dimensionless control parameters

We use the open-source MagIC numerical dynamo code[94,95] to solve the 3D magnetohydrodynamic equations in a spherical shell geometry rotating with constant angular velocity $\Omega\hat{\boldsymbol{e}}_z$. The model computes the evolution of the velocity field $\mathbf{u}$, magnetic field $\mathbf{B}$, and temperature T using the Boussinesq approximation. Details on the derivations of these governing equations and key approximations can be found at http://magic-sph.github.io/and in standard texts[96].

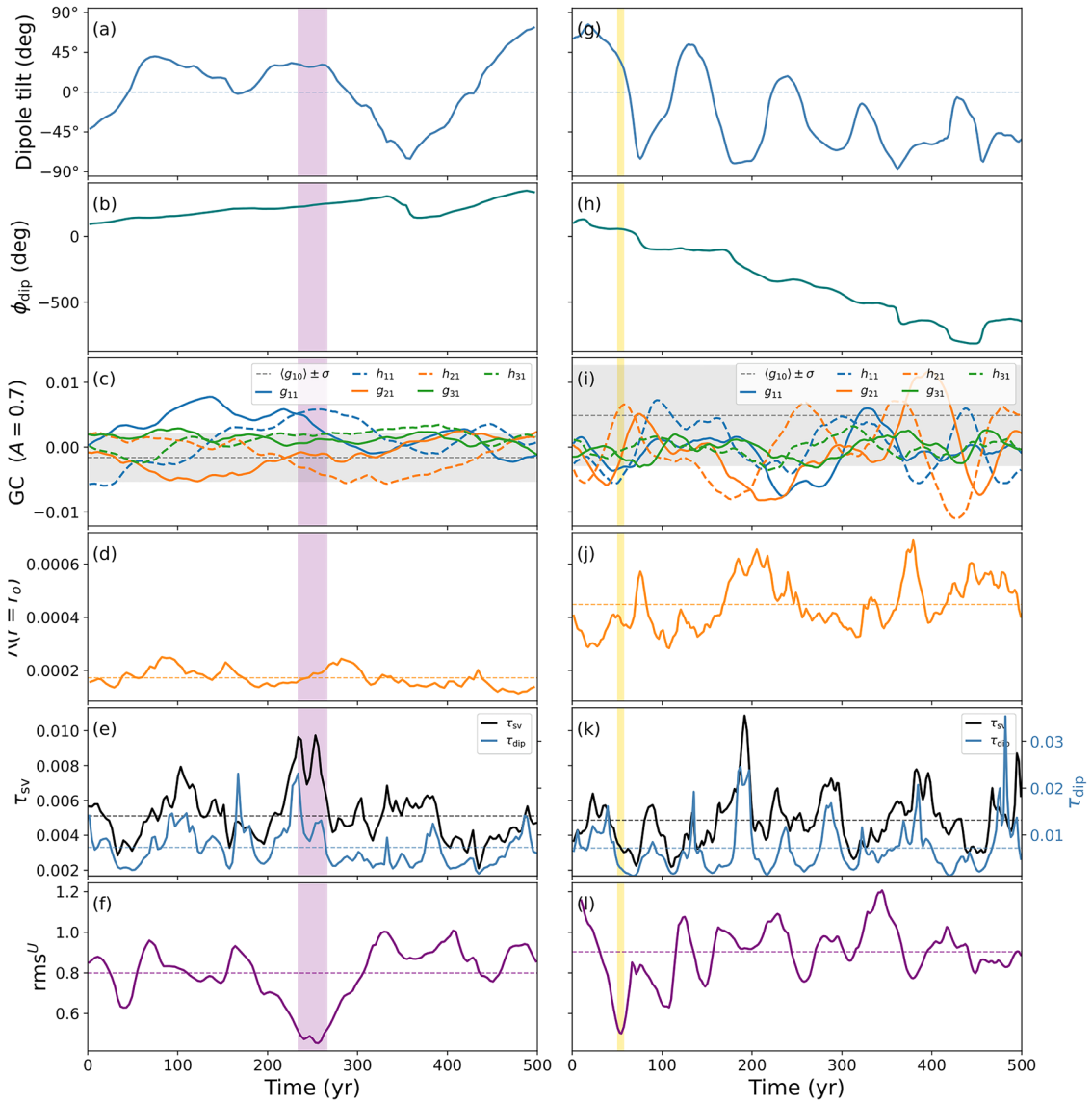


Fig. 4 | Temporal evolution of two dynamo models during their most Uranus-like intervals. a–f Model 6 ($\chi = 0.5$, σ_2); **g–l** Model 11 ($\chi = 0.6$, σ_3). Both are shown over a 500-year window and assume a dynamo depth of 0.7R. Dimensional times use the magnetic diffusion time τ_λ ("Methods"; $\sigma = 10^4$ S/m); $\tau_\lambda \approx 32$ kyr for Model 6 and ≈20 kyr for Model 11 (cf. ≈25 kyr for Model 11 in Fig. 3, where $A = 0.8$ and Neptune's radius apply), so the window spans ≈0.016 and 0.025 τ_λ, respectively. From top to bottom, panels show: **a**, **g** dipole tilt; **b**, **h** continuous dipole longitude drift ϕ_{dip}; **c**, **i** Gauss coefficients g_{11}, h_{11} (blue), g_{21}, h_{21} (orange), and g_{31}, h_{31} (green) at the planetary surface, with the gray band indicating the time mean and $\pm 1\sigma$ standard deviation of the axial dipole g_{10}; **d**, **j** Elsasser number at the top of the dynamo region, Λ; **e**, **k** secular-variation timescales τ_{SV} (black, left axis) and τ_{dip} (blue, right axis), computed using $l_{max} = 3$; and **f**, **l** morphological misfit rms^U relative to the observed Uranus field (HB96 model, $l_{max} = 3$). Shaded intervals mark snapshots where rms^U falls below 0.54, identifying the periods of closest morphological resemblance to Uranus. Dashed lines in (**d**–**f**, **j**–**l**) indicate the time-window mean. These identify how closely and how often model snapshots resemble the single observed epoch (see main text).

Using the shell thickness $D = r_o - r_i$ as a length scale (where r_o and r_i are the outer and inner boundary radii, respectively), the viscous diffusion time $\tau_\nu = D^2/\nu$ as a time scale (where ν is the kinematic viscosity), a magnetostrophic estimate $B_0 = \sqrt{\Omega\mu_0\lambda_i\rho}$ as a magnetic field scale (where μ_0 is the permeability of vacuum, λ_i is the magnetic diffusivity at the inner boundary, and ρ is the constant density in the dynamo region), and $T_{scale} = \alpha q_0 D/k$ as the temperature scale (where α is the thermal expansion coefficient, $q_0 = -k\frac{dT}{dr}\big|_{r_o}$ is the mean heat flux at the outer boundary, and k is the thermal conductivity), the non-dimensional equations become:

$$\left(\partial_t + \mathbf{u}\cdot\nabla\right)\mathbf{u} = -\nabla p - \frac{2}{E}\hat{\boldsymbol{e}}_z\times\mathbf{u} + \frac{1}{EPm_i}(\nabla\times\mathbf{B})\times\mathbf{B} + \frac{Ra}{Pr}T\frac{\mathbf{r}}{r_o} + \nabla^2\mathbf{u}, \tag{1}$$

$$\left(\partial_t - \frac{\tilde{\lambda}}{Pm_i}\nabla^2\right)\mathbf{B} = \nabla\times(\mathbf{u}\times\mathbf{B}) - \frac{\nabla\tilde{\lambda}}{Pm_i}\times(\nabla\times\mathbf{B}) \tag{2}$$

$$\nabla\cdot\mathbf{u} = 0, \tag{3}$$

$$\nabla\cdot\mathbf{B} = 0, \tag{4}$$

$$\left(\partial_t - \frac{1}{Pr}\nabla^2\right)T = -\mathbf{u}\cdot\nabla T + \epsilon \tag{5}$$

where $\hat{\boldsymbol{e}}_z$ is the unit vector along the rotational axis, p is the modified pressure including the centrifugal acceleration, $\mathbf{r}$ is the position vector,

$\tilde{\lambda}(r) = \lambda(r)/\lambda_i$ is the normalized magnetic diffusivity, and ϵ controls the uniformly distributed buoyancy sources in the dynamo region and is set to zero.

The non-dimensional control parameters in the equations are the Ekman number that represents the ratio of viscous to Coriolis forces:

$$E \equiv \frac{\nu}{\Omega D^2}, \tag{6}$$

the Prandtl number that represents the ratio of kinematic viscosity ν to thermal diffusivity κ:

$$Pr \equiv \frac{\nu}{\kappa}, \tag{7}$$

the modified magnetic Prandtl number that represents the ratio of kinematic viscosity to magnetic diffusivity λ_i at the inner boundary:

$$Pm_i \equiv \frac{\nu}{\lambda_i}, \tag{8}$$

and the Rayleigh number that represents buoyancy forcing relative to viscous and diffusive damping:

$$Ra \equiv \frac{g_o r_o^3}{\nu\kappa} T_{\text{scale}}, \tag{9}$$

where g_o is the gravitational acceleration at the outer boundary.

The magnetic diffusivity and electrical conductivity profiles are related: $\lambda(r) = 1/(\mu_0\sigma(r))$. Following Gómez-Pérez et al.[58] and others, the normalized electrical conductivity $\tilde{\sigma}(r) = \sigma(r)/\sigma_i$ in the dynamo region ($r_i \le r \le r_o$) is described by a segmented function:

$$\tilde{\sigma}(r) = 1 + (\tilde{\sigma}_m - 1)\left(\frac{r - r_i}{r_m - r_i}\right)^a \quad \text{for } r < r_m, \tag{10}$$

$$\tilde{\sigma}(r) = \tilde{\sigma}_m \exp\left[a\left(\frac{r - r_m}{r_m - r_i}\right)\frac{\tilde{\sigma}_m - 1}{\tilde{\sigma}_m}\right] \quad \text{for } r \ge r_m. \tag{11}$$

In the inner core region, $0 \le r < r_i$, the electrical conductivity is held constant at the dynamo-shell value at r_i, i.e., $\sigma(r < r_i) = \sigma(r_i)$ in all simulations. At the transition radius r_m, the electrical conductivity decreases to $\tilde{\sigma}_m$, i.e., $\tilde{\sigma}(r_m) = \tilde{\sigma}_m$. The parameter a represents the decay rate, characterizing the exponential decrease in conductivity within the outer semiconducting layer. Three non-uniform conductivity profiles are considered, set to span a range from steep to gradual decay, reflecting plausible gradients between the deep ionic and molecular layers. Figure 1 shows the radial conductivity profiles, while the adopted (r_m, σ_m, a) values are given in Table 1.

These equations require boundary conditions at both inner and outer radii of the convective spherical shell domain, as well as an initial state of fields. At both r_i and r_o, we use fixed flux boundary conditions for the temperature and stress-free boundary conditions for the velocity field. Stress-free mechanical boundary conditions are used to approximate weak viscous coupling with the molecular envelope and the deeper interior, consistent with previous ice giant dynamo studies[48,51]. We adopt stress-free rather than no-slip conditions because the accessible Ekman numbers are far larger than planetary values, so no-slip boundaries would impose artificially strong viscous Ekman boundary layers. The choice of mechanical boundary condition can nonetheless influence the geostrophic zonal flows and field morphology of dynamo models[67,97–100]. For the magnetic field, we use an electrically insulating outer and a finitely conducting inner boundary.

The inner region ($r < r_i$) is modeled as a freely rotating solid whose rotation rate evolves self-consistently in response to the magnetic (Lorentz) torque, maintaining magnetic coupling between the inner region and the dynamo shell. Fluid motion within this region is not modeled, as it is treated as a solid representing the deep interior beneath the convecting shell—possibly superionic ice or compositionally stratified material—whose rheology and dynamical state remain observationally unconstrained. This treatment is well established in numerical dynamo studies[101], and the magnetic morphologies reported here depend primarily on the radial conductivity gradient across the convecting shell rather than on inner-core dynamics.

We use a pseudospectral method with 65 Chebyshev polynomials in the radial direction. Scalars on each spherical surface are resolved in latitude and longitude using SHs with maximum degree $l_{\max} = 170$. Each numerical simulation has been run for at least a few viscous diffusion times after equilibrium. Time-averaged diagnostic values in Table 2 are computed over one viscous diffusion time (0.2 magnetic diffusion times) after the dynamo models have equilibrated, and all magnetic spectra are time-averaged over half a viscous diffusion time (0.1 magnetic diffusion times).

Diagnostic parameters

Key output diagnostics of the bulk fluid (given in Table 2) include the magnetic Reynolds number (ratio of magnetic induction to magnetic diffusion):

$$Rm = \mu_0\sigma_i UD, \tag{12}$$

where U is the characteristic flow speed, the Elsasser number (ratio of Lorentz to Coriolis forces):

$$\Lambda = \frac{\sigma B^2}{\Omega\rho} \tag{13}$$

where B is the characteristic magnetic field intensity, and the local Rossby number (ratio of inertial to Coriolis forces on the convective length scale):

$$Ro_l = Ro\frac{\bar{l}_{\mathbf{u}}}{\pi}, \tag{14}$$

where Rossby number $Ro = U/(\Omega D)$, $\pi/\bar{l}_{\mathbf{u}}$ is the characteristic half-wavelength of the flow and $\bar{l}_{\mathbf{u}}$ is derived from the time-averaged kinetic energy spectrum,

$$\bar{l}_{\mathbf{u}} = \frac{\sum l\langle \mathbf{u}_l \cdot \mathbf{u}_l\rangle}{2E_{\text{kin}}}, \tag{15}$$

with E_{kin} being the kinetic energy integrated in the dynamo volume[102].

We define f_{M1} as the ratio of surface magnetic power in the SH order $m = 1$ to the axial component $m = 0$:

$$f_{\text{M1}} = \frac{\mathcal{P}_{m=1}}{\mathcal{P}_{m=0}} = \frac{\sum_{l\ge1}^{l_{\max}}\mathcal{P}_{l,m=1}}{\sum_{l\ge1}^{l_{\max}}\mathcal{P}_{l,m=0}}, \tag{16}$$

where $l_{\max} = 6$ for the simulations and the maximum resolved degree for each observed field model ($l_{\max} = 3$ for HB96 and O_8, $l_{\max} = 4$ for AH_5).

We additionally report the following snapshot statistics:

1. p_{tilt} is the fraction of snapshots for which the dipole tilt lies within ± 10° of the *Voyager 2* values (59° for Uranus and 47° for Neptune).
2. $p_{M>1}$ is the fraction of snapshots for which the total magnetic power in spherical-harmonic order $m = 1$, summed over all degrees $l \ge m$, exceeds the axisymmetric power in $m = 0$.
3. $p_{l>1}$ is the fraction of snapshots for which the power in either degree $l = 2$ or $l = 3$ exceeds the dipole power at $l = 1$.

Snapshot-based morphology diagnostics

To quantitatively compare the dynamo models with the observed magnetic morphology of Uranus and Neptune, we define two complementary scores.

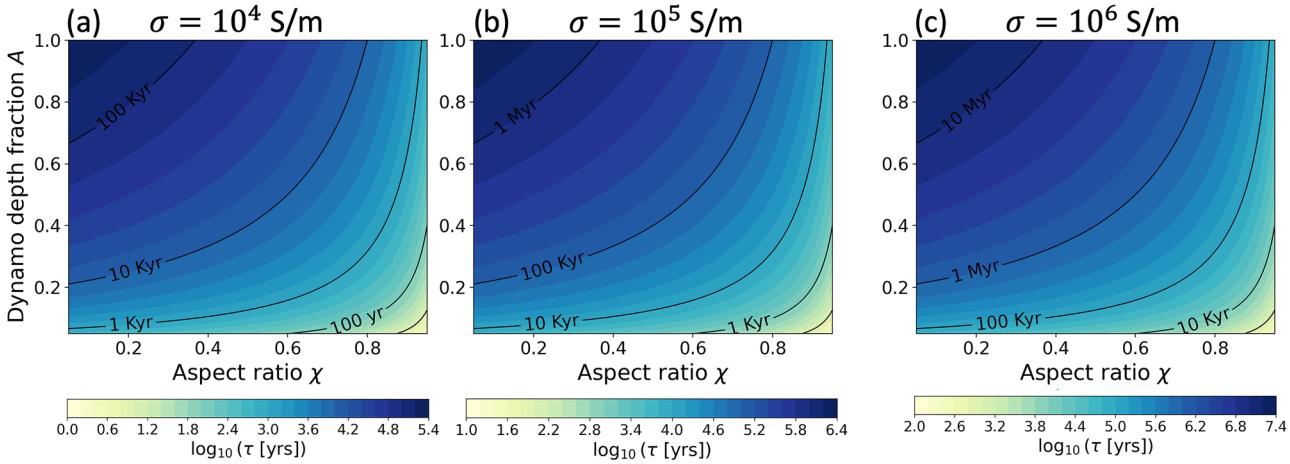


Fig. 5 | Dimensional magnetic diffusion time. τ_λ as a function of aspect ratio χ, dynamo depth A, and electrical conductivity σ. Each panel shows $\log_{10}$ of $\tau_\lambda = \mu_0 \sigma D^2$ (in years), with $D = (1-\chi)AR$, as a function of χ and A, for electrical conductivities of **a** $\sigma = 10^4$ S/m, **b** $\sigma = 10^5$ S/m, and **c** $\sigma = 10^6$ S/m. The planetary radius is approximated as $R \approx 2.5 \times 10^7$ m. Black contours indicate values of τ_λ in years.

The first score evaluates how closely individual snapshots of a modeled surface field resemble Uranus' or Neptune's as observed during the *Voyager 2*-era, while the second score evaluates whether the time-averaged model field reproduces the observed dominance of the spectral M1 peak. These two diagnostics emphasize different aspects of the magnetic morphology: the snapshot-based RMS score measures the occurrence and similarity of instantaneous Uranus- or Neptune-like fields, whereas the M1 score measures the long-term tendency of a model to maintain the observed M1 peak distribution.

For each model X, we first define a morphology score $S^{(p)}_{\mathrm{rms},X}$ from the RMS mismatch values of its snapshots. For a given planet p, where p = U for Uranus or p = N for Neptune, the morphology score is

$$S^{(p)}_{\mathrm{rms},X} = -\ln\left[\frac{1}{N_X}\sum_{i\in\mathcal{T}_X}\exp\left(-\frac{1}{2}\left[\frac{\mathrm{rms}^{(p)}_i}{\sigma_{\mathrm{rms}}}\right]^2\right)\right], \tag{17}$$

where $\mathrm{rms}^{(p)}_i$ is the RMS mismatch between snapshot i and the observed magnetic spectrum of planet p, and $\mathcal{T}_X$ denotes the snapshots from model X that fall among the 100 lowest-RMS snapshots pooled across all models. The sum therefore runs over model X's contributions to this global set of best matches, while the normalization uses N_X, the total number of snapshots from model X. In practice, each model is sampled at a comparable cadence over a comparable post-equilibration interval, yielding ~1000 snapshots per model, so the normalization by N_X accounts for any residual differences in snapshot count. This means that a model scores well only if it produces low-RMS snapshots frequently, not merely occasionally. The parameter σ_{rms} = 0.2 sets the RMS tolerance scale below which snapshots are considered good matches.

Here, $\mathrm{rms}^{(p)}_i$ is defined as

$$\mathrm{rms}^{(p)}_i = \min_{\epsilon=\pm1,\,\phi_0} \mathrm{rms}\left(\epsilon\tilde{\mathbf{B}}_{\mathrm{X},i}(\phi_0),\ \tilde{\mathbf{B}}_{\mathrm{obs,p}}\right) \tag{18}$$

where the modeled field $\tilde{\mathbf{B}}_{\mathrm{X,i}}$ and observed field $\tilde{\mathbf{B}}_{\mathrm{obs,p}}$ are each normalized by their own RMS amplitude. This normalization focuses the comparison on morphology rather than field intensity. Field morphologies are compared, allowing both magnetic polarities ($\epsilon = \pm1$) and arbitrary azimuthal rotations (ϕ_0) since the simulations do not prescribe an absolute magnetic polarity or longitude. For Uranus' and Neptune's observed fields, both are truncated at $l_{\max} = 3$, because single *Voyager 2* flybys robustly constrain only the low-degree dipole, quadrupole, and octupole fields. The published inversions extend to l = 3 for HB96 and O_8, and to l = 4 for AH_5. The modeled snapshots are correspondingly truncated to $l_{\max} = 3$ before the RMS comparison, so the score reflects only the robustly resolved large-scale morphology and discards smaller-scale $l \geq 4$ structure that the observations do not constrain. The modeled magnetic fields are evaluated at the planetary surface assuming the top of the dynamo region to be at A = 0.7, 0.75, and 0.8. These values are representative assumed dynamo depths used to test the effect of geometric attenuation through the overlying molecular envelope; they are not independently fitted planetary interior parameters.

Models 12–14 serve only to test the robustness of the M1 peak to different E and Ra values and are excluded from the ranking. For each magnetic field model and estimate of the dynamo surface, we rank the top 100 most Uranus-like and 100 most Neptune-like snapshots by the global minimum $\mathrm{rms}^{(p)}$ (e.g., Supplementary Fig. 12).

We then define a second score based on the time-averaged M1 dominance, $\overline{f}_{\mathrm{M1},X}$, of each model X. The observed values are

$$f^{(p)}_{\mathrm{M1,obs}} = \begin{cases} 2.3, & p = \mathrm{Uranus(AH_5[6])}, \\ 2.1, & p = \mathrm{Neptune(\ HB96[5])}, \end{cases} \tag{19}$$

as shown in Supplementary Note 1. Because f_{M1} is a ratio of magnetic powers, we compare the model and observed values using a logarithmic mismatch. The M1 score is

$$S^{(p)}_{\mathrm{M1},X} = \frac{1}{2}\left[\frac{\ln\left(\overline{f}_{\mathrm{M1},X}/f^{(p)}_{\mathrm{M1,obs}}\right)}{\sigma_{\ln f}}\right]^2. \tag{20}$$

This form penalizes fractional deviations from the observed M1 dominance symmetrically: a model that overestimates or underestimates f_{M1} by the same factor receives equal penalty. We take $\sigma_{\ln f} = \ln(1.5)$, corresponding to a factor of 1.5 tolerance in the M1 dominance. This choice provides a moderate penalty for models with slightly stronger or weaker M1 peaks than observed, while strongly penalizing models that fail to produce substantial m = 1 dominance.

The total score for model X and planet p is then defined as

$$S^{(p)}_{\mathrm{tot},X} = S^{(p)}_{\mathrm{rms},X} + \lambda S^{(p)}_{\mathrm{M1},X}, \tag{21}$$

where λ controls the relative importance of the time-averaged M1 dominance compared with the snapshot-based morphology score. We set λ = 1 so that both terms contribute equally. Lower values of S_{tot} indicate better agreement with the observed magnetic morphology. This combined score therefore favors models that both frequently generate low-RMS

Table 3 | Magnetic diffusion time dimensionalization

Planet	A	Models 0–3 $\chi = 0.35$	Models 4–7 $\chi = 0.5$	Models 8–11 $\chi = 0.6$
	0.7	54	32	20
Uranus	0.75	62	37	23
	0.8	71	42	27
	0.7	50	30	19
Neptune	0.75	57	34	22
	0.8	65	39	25

Dimensional magnetic diffusion time $\tau_\lambda = \mu_0 \sigma D^2$, with shell thickness $D = (1-\chi)AR$, at dynamo depths $A = 0.7, 0.75, 0.8$. Models sharing an aspect ratio are grouped by column. Values assume bulk $\sigma = 10^4$ S/m and radii $R_U = 2.56 \times 10^7$ m (Uranus) and $R_N = 2.46 \times 10^7$ m (Neptune). All τ_λ values are in kyr.

Uranus- or Neptune-like snapshots and maintain a time-averaged M1 dominance close to the observed value.

Because $S_{M1} < 0.5$ for the top-ranked models of both planets at all dynamo depths (see Supplementary Tables 2 and 3), the combined score S_{tot} is insensitive to the exact choice of the M1 tolerance parameter $\sigma_{\ln f}$: varying it within reasonable bounds does not change which models rank in the top three as discussed in the main text. The ranking is therefore effectively determined by snapshot morphology alone for the best-performing models, with S_{M1} serving primarily to eliminate models that fail to produce the required non-axisymmetric field on average. Likewise, varying the morphology tolerance σ_{rms} over [0.1, 0.3] and the number of best-matching snapshots over [50, 200] leaves the set of top-performing models—and the conclusion that thin-shell variable-conductivity models most closely resemble the observed fields—unchanged, although the precise rank order within this set can shift, consistent with the relative-order variability noted in the main text.

Temporal variability diagnostics

The secular variation timescales are calculated from the Mauersberger–Lowes RMS radial field strength B_r and its instantaneous rate of change $\dot{B}_r$. We compute three versions of this diagnostic: τ_{SV} for the full field (evaluated at $l_{max} = 3$, matching the truncation of the secular-variation comparisons), τ_{dip} for the dipole component only, and τ_{ndip} for the non-dipolar components, here including degrees $2 \le l \le 3$:

$$\tau_{SV} = \frac{B_{r,tot}}{\dot{B}_{r,tot}}, \qquad \tau_{dip} = \frac{B_{r,dip}}{\dot{B}_{r,dip}}, \qquad \tau_{ndip} = \frac{B_{r,ndip}}{\dot{B}_{r,ndip}}, \tag{22}$$

In dimensional units, the magnetic diffusion time—the timescale over which magnetic fields evolve—may be significantly shorter in Uranus and Neptune than in dipole-dominated planetary dynamos: $\tau_\lambda = \mu_0 \sigma D^2$, where $D = r_o - r_i = (1-\chi)AR$ is the dynamo shell thickness and $\sigma = (\mu_0 \lambda)^{-1}$ the electrical conductivity. Given their comparatively low electrical conductivity and potentially thin convective shells, τ_λ is expected to be short in the ice giants.

The inferred secular-variation timescale therefore scales as $\tau_\lambda \propto \sigma D^2$. An alternative is the magnetic advective timescale $\tau_U = \tau_\lambda / Rm = D/U$, which relates secular variation to the convective velocity U through the magnetic Reynolds number Rm; dynamo scaling laws relate U, and hence Rm, to the available buoyancy flux and rotation[67]. Because U—equivalently the planetary magnetic Reynolds number—is not directly constrained for the ice-giant dynamo regions, we retain τ_λ as our baseline. Secular variation is also degree-dependent, with higher spherical-harmonic degrees evolving more rapidly ($\tau_{SV} \propto 1/l^{12}$), consistent with our degree-resolved diagnostics, in which the non-dipolar timescale τ_{ndip} is systematically shorter than the dipole timescale τ_{dip}.

Figure 5 plots the dimensional magnetic diffusion time as a function of the dynamo region's aspect ratio (χ) and dynamo depth (A), with electrical conductivities ranging between $\sigma = 10^4$–10^6 S/m (Table 3). For example, assuming a dynamo shell with aspect ratio $\chi = 0.35$, a dynamo surface at $0.75R$ with $R \approx 2.5 \times 10^7$ m, and electrical conductivity $\sigma = 10^4$ S/m for Uranus and Neptune[103], the estimated magnetic diffusion time is approximately 59 kyr. For a thinner shell geometry of $\chi = 0.6$ and other conditions unchanged, the estimated magnetic diffusion time drops to 22 kyr. If the electrical conductivity is comparable to Earth's core (i.e., $\sigma = 10^6$ S/m), then the magnetic diffusion time increases by two orders of magnitude.

For re-dimensionalization of the ice-giant-like cases, we adopt electrical conductivity $\sigma = 10^4$ S/m, aspect ratios $\chi = 0.5$–0.6, dynamo depths $A = 0.7$–0.8, and planet radii $R_U = 2.56 \times 10^7$ m for Uranus and $R_N = 2.46 \times 10^7$ m for Neptune. As an example, for Model 6, the dimensional magnetic diffusion time is estimated to be 32 kyr, with aspect ratio $\chi = 0.5$, dynamo depth $A = 0.7$, and $R = R_U$. For Model 11, the dimensional magnetic diffusion time is estimated to be 19 kyr, with aspect ratio $\chi = 0.6$, dynamo depth $A = 0.7$, and $R = R_N$. These dimensional estimates should be regarded as order-of-magnitude illustrative values owing to uncertainties in the electrical conductivity structure and dynamo shell geometry of Uranus and Neptune (Table 3).

Data availability

The data supporting the findings of this study—the simulation control files, magnetic Gauss coefficient time series, and time-averaged magnetic energy spectra for all 15 dynamo models, the observed-field coefficient files, and the derived morphology-misfit and secular-variation diagnostics—are available in the figshare repository at https://doi.org/10.6084/m9.figshare.32447910. The source data for Figs. 2–5 are provided with this paper as Supplementary Data 1–4, respectively. Figure 1 is schematic: the electrical conductivity profiles shown are fully specified by the analytic expressions and parameters given in the "Methods" section (Table 1), and the interior model points shown for comparison are from ref. 33. The observed magnetic field models are from the cited publications (AH_5[6], HB96[5], and O_8[4]).

Code availability

The numerical dynamo code MagIC is open source (https://magic-sph.github.io/; version 6.3, git v6.3-96-g8bdb98b, used here) and is built on the SHTns library (https://bitbucket.org/nschaeff/shtns/). The analysis and figure-generation scripts (Python 3, using the MagIC Python tools, numpy, matplotlib, and openpyxl), including the scripts that generate Supplementary Data 1–4, are archived together with the dataset in the figshare deposit at https://doi.org/10.6084/m9.figshare.32447910.

Received: 6 February 2026; Accepted: 18 August 2026;
Published online: 03 November 2026

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Acknowledgements

We thank Mandy Bethkenhagen, Federica Coppari, Sebastien Hamel, Koutaro Hikosaka, Weijia Kuang, Marius Millot, Yoshiyuki Okuda, Sabine Stanley, and Allona Vazan for helpful discussions. This work used the Stampede3 system at the Texas Advanced Computing Center (TACC) through allocation PHY240217 from the Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support (ACCESS) program[104], which is supported by U.S. National Science Foundation grants #2138259, #2138286, #2138307, #2137603, and #2138296.

Author contributions

C.Y. and K.M.S. conceived the study. C.Y. performed the simulations and analysis, curated the data, and wrote the original draft. K.M.S. acquired funding, supervised the project, and revised the manuscript. J.M.A. and H.C. contributed to the interpretation of the dynamo results. A.M. contributed to developing and writing the discussion of the magnetospheric implications. H.M. contributed to the interpretation of auroral observations. All authors approved the final manuscript.

Funding

C.Y. and K.M.S. were supported by the University of Texas Institute for Geophysics (UTIG) in the Jackson School of Geosciences. C.Y. received additional support from NASA Cassini Data Analysis Program grant 80NSSC21K0529, and K.M.S. from NASA grant NNX15AL56G. H. M. was supported by an STFC James Webb Fellowship (ST/W001527/2) at Northumbria University, UK.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s42005-026-02841-y.

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Peer review information *Communications Physics* thanks Tom Stallard and the other anonymous reviewers for their contribution to the peer review of this work. A peer review file is available.

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